A Scientific Discussion and Critical Analysis of the paper entitled - A New Set of Cluster-Driven Composite Development Indicators

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Abstract

The selection of what and how indicators should be aggregated to form composite development indicators is an essential topic of debate. Although methods exist in the literature to aid the process, such as principal component analysis (PCA). It is shown that they may ignore critical information on the relationship between indicators. This paper provides a scientific discussion and critical analysis of the methods presented in the paper - A new set of cluster driven composite development indicators. Advancements in network filtering techniques are applied to form a new set of cluster-driven composite development indicators that are objective, data-driven and economically intuitive [1]. This paper discusses their results and financial interpretations and compares them with the top PageRank indicators and a random benchmark. It was found that cluster-driven composite indicators outperform both. Finally, an improvement is discussed based on the underlying assumptions of the methodology applied.

This paper has been formatted using the guidelines specified in the research methods booklet, namely 11pt font and 2cm margins.

1 Introduction

Composite development indicators consolidate multiple individual indicators into a singular number that aims to provide a holistic assessment of the level of development of a country [2]. The primary purpose of composite indicators is to monitor a country's development and evaluate its advancement over time [1].. An example is the Human Development Index (HDI) [3]. As the number of indicators is ever-increasing, this feature is highly desirable to inform policymakers of the economic, social and environmental development due to the challenges associated with interpreting high dimensional data [1]. However, composite development indicators must be cautiously interpreted as they suffer various drawbacks. Firstly, they are an oversimplification of a complex system that resembles the collective dynamics of interconnected variables [4]. Secondly, the comparability of indicators must be considered, as countries may define indicators differently. For example, the definition of "looking for work" varies by country [5]. Finally, their validity is inherently tied to how they were constructed [2]. A composite indices review paper by S. Greco et al. identifies two highly criticised issues, namely their weighting and aggregation [2]. The process of weighting and aggregating individual indicators involves some degree of subjective judgement of what constitutes development due to a lack of a theoretical framework [2]. As a result, biases may be introduced with current approaches to construction, and some information may be missed or overemphasised [1]. Aggregating indicators is also seen as a trade-off between complexity and the loss of information [2].

Given how consequential these indicators are due to their role in characterising development and forming policy decisions, they deserve social-scientific attention [5]. The paper entitled "A new set of cluster-driven composite development indicators" aims to tackle some of the drawbacks described above by providing guiding principles on how composite development indicators should be aggregated. The construction of a meaningful composite development indicator is detailed by utilising advancements in the field of Econophysics, namely network filtering techniques and the relation between indicators to identify in an objective and data-driven way the vital indicators characterising a country's development [1]. The aforementioned paper will be the focus of this scientific discussion and critical analysis.

The paper uses a large dataset of indicators and first looks at an existing data-driven dimensionality reduction technique called Principal Component Analysis (PCA), which is commonly used in composite indicators produced by large organisations [2]. PCA aims to capture the highest possible variance while reducing the number of input variables in a dataset[2]. However, the paper finds that while this method offers an insight into the structure of the data, it may not be suitable for forming a new composite development indicator [1]. Because this may discard relevant information [1]. It is also found that the preassigned topics are not the best way to aggregate the indicators [1]. Following this, the paper then turns to the aid of network filtering techniques and hierarchical clustering algorithms commonly used in the field of Econophysics [1]. This paper is the first time methods such as the Planar Maximally Filtered Graph (PMFG) and Directed Bubble Hierarchical Tree (DBHT) are used to analyse the structure of indicators [1]. The results reveal that the clusters of indicators are economically comprehensible and contain a mixture of topics [1]. The results are then applied, forming the new method of how/what indicators to aggregate, called Cluster Driven Composite Indicators (CDCI) [1]. Another method to form composite indicators using he most influential indicators are subsequently found using PageRank. The performance of both methods are assessed against a random benchmark, finding that the newly discovered CDCIs outperforms its counterparts.

The following academic discussion on the paper is organised in the following way. Section 2 further discusses the existing approaches used to aggregate composite indicators. Focus is then turned to the PCA and the field of Econophysics, where the background and methodology behind the proposed composite indicator are discussed. Section 3 looks at the results and their economic interpretation. Finally, an improvement to the method is suggested at the end of section 3, followed by conclusions.

2 Background and Methodology

The process of constructing a composite development indicator is essential. The weighting applies explicit importance to each indicator, directly impacting the overall composite indicator [6]. The weights have been applied in the literature, ranging from subjective to objective approaches [2]. The selection of how and what indicators are aggregated has also been an important topic of debate. Strategies have ranged from using theory, empirical evidence and analysis, public consensus, and even intuitive appeal [6]. As a result, it is commonplace for researchers to face criticism for their selection of indicators and dimensions [6]. Subjective methods suffer from potential bias [2]. However, methods exist in the literature to aid in selecting indicators such as PCA or FA [2]. While PCA is more often used to elicit or compare the weights for the indicators [2]. A paper by Ogwang and Abdou reviews the ability to use PCA/FA to select principal variables for computing measures of human well-being [7]. It is found that these methods may be used to choose a subset of variables to include in the construction of composite indicators that adequately explain the overall variation of the dataset [7]. However, another paper highlights the challenge of creating composite indicators through dimensionality reduction techniques such as PCA, as they may disregard important information. This paper contributes to the literature review by adding a new objective and data-driven method to composite indicators that aim to overcome some of the complications faced by existing methods. However, it is essential to exercise caution when constructing indicators using an objective data-driven approach as correlations between indicators may not consistently reflect the true causal influence between them, resulting in conceptually meaningless relations [2].

Previous papers related to the field have applied hierarchical clustering to analyse clusters of countries [8][9]. However, this paper is the first time hierarchical clustering algorithms have been used to look at the structure of indicators and develop new indicators. In the broader field, network filtering and hierarchical clustering methods have been applied to look at the industrial sector classification of stocks [10]. More specifically, the PMFG and DBHT later described have been applied to gene expression patterns of lymphoma, successfully identifying significant groups of genes used for diagnosis [11]. As the paper of discussion is relatively new, no papers directly apply the methodology presented. However, the methodology has been referred to as an improvement that goes beyond the scope of another paper [12]. The background of this methodology is discussed in detail in the next section.

This remainder of this section aims to provide a comprehensive overview of the background of network filtering techniques and hierarchical clustering, setting the stage for the subsequent analysis and discussion. To begin, this paper looks at the dataset and the methodology used to interpret its structure. The paper draws data from the World Development Indicator (WDI) dataset, which contains a large number of yearly development indicators for $C = 218$ countries (C represents the total number of countries) [1]. Even though the data is taken from an officially recognised agency, it must be noted that the dataset suffers from very high levels of missing data, as mentioned in the supplementary [1]. This is particularly important as the data is not missing at random, and the dataset is biased towards more developed countries [1]. While this has been treated using an appropriate imputation scheme to preserve the correlation structure, it must be recognised that predictions should not be made using the data [1].

The paper focuses on data for $T = 19$ years, where T represent each consecutive year from 1998 to 2016, as these years possess the lowest amount of missing data [1]. The paper looks at a total of $N = 1448$ indicators across each of these years after removing trivially correlated indicators from the dataset[1]. The dataset divides these 1448 indicators into 94 different topics [1]. The paper further amalgamates these into $G = 12$ different topics based on root objectives to increase the statistical reliability of any conclusions. The different topics will be used to assess the structure of the data following the application of the techniques described later.

The paper examines the structure of the data by analysing the correlations across multiple years, aiming to quantify the relationships between different indicators [1]. A data matrix X is constructed by vertically

stacking C matrices of size T $\times$ N, where each cross-sectional block represents a particular country c, and each column reports the data for the corresponding indicator i [1]. The rows correspond to the year; the first row is the first year t=1, and the last is the last year t=T [1]. When finding the overall data structure, removing trends and eliminating spurious correlations is crucial [1]. To do so, the first difference $\Delta X(\tilde{t}, c, i)$ is calculated for each block of data using the formula:

$$\Delta X(\tilde{t}, c, i) = X(t+1, c, i) - X(t, c, i) \tag{1}$$

In the above formula, X(t,c,i) refers to the indicator i's value for country c in year t and X(t+1,c,i) is the same but for the next year [1]. The first difference is calculated for $\tilde{t}$ from 1 to 18 = (T-1). After stacking vertically every $\Delta X(., c, .)$ of size T-1 x N, the matrix ΔX is formed [1]. The relationship between the indicators is then encoded using the empirical Pearson correlation matrix E to quantify the strength and direction of associations between indicators. Calculated from a zero mean, standardised ΔX as

$$E = \frac{1}{C(T-1)} (\Delta X)^\top X \tag{2}$$

where $\top$ is the transpose. From the structure of E, we aim to assess the correlation between development indicators. However, given that the size of the raw correlation matrix is 1448 x 1448 and inherent noise in the system, it may retain redundant information and is not optimal to use [1]. This paper now discusses two methods to not only deal with the problem but also enhance the interpretability of E's structure.

Dimensionality reduction methods are techniques used to reduce the number of input variables in a dataset while preserving the most important information. PCA is the most commonly used dimensionality reduction technique on correlation matrices and is often chosen due to its ease of use [2]. For these reasons, the method is only briefly described. PCA identifies a subset of orthogonal basis vectors (principal components) along which the data varies the most for the correlation matrix E [1]. The eigenvectors with the highest eigenvalues capture the most variance in the data [13]. These eigenvectors are the principal components, and the orthogonality property ensures that each principal component captures an independent source of variability. Each subsequent principal component explains a diminishing proportion of the total variation in the data [1]. By selecting a subset of these principal components, we can effectively reduce the dimensionality of the data while retaining the most significant sources of variation [1]. However, PCA is sensitive to the presence of outliers [13]. Economic indicators such as GDP or income are often characterised by a power-law distribution, meaning that a few countries contribute disproportionately, and outliers may exist in the data [14]. However, as the data has been standardised and normalised, there is no need to apply robust variants of PCA in this paper[13]. This paper looks at PCA to form a composite indicator in a data-driven way by using the subset of principal components as a basis. However, PCA retains a subjective component in deciding the subset of principal components. While Random Matrix Theory (RMT) can aid in deciding, PCA may not be the best method to achieve the goal of this paper[1].

The Eigenvectors are investigated to interpret the contributions of the G amalgamated topics to the principal components to reveal their structure. As the principal components explain a decreasing total variation of the data, looking at the first few will help identify the topics most significantly contributing to development [1]. The structure is revealed by projecting each eigenvector v_i in E onto the G topics [1]. The projection is performed using a matrix P, where each entry $P_{ig} = 1/N_g$ if i is in topic g, and 0 otherwise. Here N_g is the total number of indicators in the individual topic g. The paper now introduces a G-dimensional vector ρ_i for each eigenvector v_i, comprising of entries $\rho_{g,i}$ signifying the contribution of the gth topic to the ith eigenvector [1]. Computing ρ_i by the formula,

$$\rho_i = \gamma_i P v_i, \tag{3}$$

where γ_i is a normalisation constant $\frac{1}{2}\sum_{g=1}^{G}\rho_{g,i}$ [1]. This paper later looks at the results for when ρ_i is plotted for the top 6 principal components.

This paper details network filtering techniques and hierarchical clustering to highlight why these methods may be advantageous when forming a composite development indicator. A network, also known as a graph, is a pair G(V, E) comprised of the vertex set, V, containing vertices, v, which are connected by elements of the edge set, E, containing edges, which represent interactions between elements within complex systems [15]. A weighted graph can be constructed to depict the relative dependency and mutual information shared among different elements within a system composed of n elements [15]. Each vertex in the graph represents an element, and edges are established based on a similarity measure between pairs of elements, such as the correlation coefficient matrix of the system E [15]. Here, the correlation coefficient serves as the weight of the edge, signifying the strength of the connection between vertices. This framework allows us to model complex systems, enabling the analysis of patterns, structures, and dynamics within these systems.

One of the most basic graphs is the complete graph K_n, which contains no loops or multiple edges, and every pair of distinct vertices is connected by an edge [16]. A loop is defined as an edge of the form (v, v) which connects a vertex to itself [17]. Multiple edges are defined as when two or more edges are connected to the same two vertices [17]. The complete graph is given by the raw correlation matrix described earlier, which is where everything is connected to everything. However, the complete graph contains $N(N-1)/2$ edges, where N refers to the number of vertices [16]. This means that its efficiency diminishes in large-scale systems. The complete graph contains much redundant information, as not all pairs of vertices have meaningful interactions. In addition, it faces computational and storage challenges. Therefore, there is a need to apply network filtering techniques for correlation-based graphs to identify the key information by filtering the information into a more relevant subgraph [15].

Network filtering techniques enforce sparsity to remove noise from the correlation matrix and find connections between vertices that actually exist, thus making the hidden structure more visible [1]. This approach is favoured over thresholding, as the latter tends to subjectively cut information from complex phenomena that often encompass multi-scaling signals that cannot be trivially thresholded [11]. One of the earliest network filtering techniques was the Minimum Spanning Tree (MST), which extracts the smallest set of relevant interactions while maximising the weight of the associated correlations [15]. When constructing both the MST and the subsequently introduced Planar Maximally Filtered Graph (PMFG), the initial step involves encoding the similarity matrix E to generate a distance matrix D_{ij} [15]. The transformation satisfies properties such as non-negativity, symmetry, and the triangle inequality required for the algorithm to work [15]. Here,

$$D_{ij} = \sqrt{(2(1 - E_{ij}))}, \tag{4}$$

and the corresponding distances, which measure how far two indicators are in terms of their correlation, are minimised [1]. Hence the name MST. One of the most efficient algorithms to construct the MST is the EPMST [18]. However, the assumption that the extracted network is a tree is stringent and, therefore, loses valuable information [15]. The graph only retains N-1 edges [16]. For example, if three indicators involve similar features, only two can be connected directly via an edge in the MST [16]. Otherwise, a 3-clique is formed, failing the tree's requirements to contain no cycles [16]. A clique is defined as "a complete subgraph that links all elements"[19]. Cliques signify highly interconnected and cohesive substructures within the network and are fundamental to identifying clusters of indicators potentially sharing similar characteristics [16].

A later developed network filtering technique proposed a solution that involved building graphs under the constraint that it can be embedded on a surface with a given genus g=k [19]. The graphs produced contain

the same hierarchical tree as in the MST but also have the advantage of containing extra information about the underlying structure through loops and cliques. [19]. The information increases as the genus increases, although it is found that the greatest relative improvement on the MST is for the planar case, k=0, which is a triangulation of a topological sphere [19]. This graph is known as the Planar Maximally Filtered Graph (PMFG) and contains 3(N-2) edges [19]. The PMFG is constructed by first constructing an ordered list of edges i,j ranked by decreasing correlation [19]. Then, starting with the strongest correlation, an edge is added to the graph between elements i and j, on the condition that the resulting graph is still able to be embedded on a surface of genus g=0 [19]. This process is iterated for all possible pairs i and j. In this way, the PMFG algorithm retains only the strongest and most significant edges while eliminating weaker ones [19]. The resulting PMFG graph is planar, meaning it can be drawn on a flat surface such that no edges intersect [19]. For g=0, only cliques of three or four elements are possible according to Kuratowski's theorem; larger cliques require a larger genus but will no longer be planar [19]. Table 1 illustrates the advantages of employing network filtering techniques where N=1448 and shows the additional information retained by the PMFG over the MST.

Method	Formula	Number of Edges
Complete Graph	$(N(N-1))/2$	1047628
MST	$N-1$	1447
PMFG	$3(N-2)$	4338

Table 1: Comparison of the number of edges for different Network Filtering Techniques, N = 1448.

Hierarchical clustering algorithms are applied to group together indicators with similar properties. It is essential to look at these clusters hierarchically so that relevant local and global features are preserved [11]. The paper aims to analyze the relationships between indicators in a data-driven manner and opts for the Directed Bubble Hierarchical Tree (DBHT) method due to its compelling characteristics. The DBHT approach allows for an interpretation and partition of the indicators that align with the data while making minimal assumptions about its structure [1]. Among the various hierarchical clustering techniques in the literature, the DBHT stands out as the superior choice. Unlike many stochastic tools that yield multiple outcomes, DBHT is deterministic, ensuring reproducibility [11]. Furthermore, DBHT does not necessitate prior assumptions, such as fixing the number of clusters beforehand, unlike single linkage and k-medoids [10]. Additionally, it operates without the need for thresholding, therefore allowing the information from all indicators to be considered [11]. The DBHT was found to outperform other clustering methods, offering a robust and versatile approach to clustering analysis when applied to the field of finance [10].

The DBHT leverages the topological structure of PMFGs to extract clusters and hierarchies [11]. In this case, the PMFG is a graph G(V,E,W,D) where we additionally define W, the weight edge set using the similarity matrix E_{ij} and D, the edge-distance set using the dissimilarity matrix D_{ij}. Firstly, a hierarchy can be established due to the existence of separating 3-cliques, k_p in the structure of the PMFG. A 3-clique is so-called separating if its removal splits the graph into two disjoint and non-empty sets of vertices [11]. These sets of vertices are named the interior G_p^{in} and exterior G_p^{in} are therefore connected by the separating 3-clique [11]. After considering all the separating 3-cliques, the graph can be subdivided into bubbles, which are connected by the separating 3-cliques forming a bubble tree, denoted H_b [11]. Each vertex b_i represents the bubbles connected by the edge b_ib_j corresponding to a separating 3-clique [11].

To determine the direction of each of the edges b_ib_j in the bubble tree H_b, the sum of the weights of connections to the interior and exterior subgraphs from each separating 3-cliques k_p are compared [11]. This comparison dictates the direction of the edge towards the side with the largest weight using the formula below

$$W_p^{in/ex} = \sum_{v \in k_p, u \in G_p^{in/ex}} A_G(v,u) \tag{5}$$

where $A_G(v,u) = w_{vu}$ is the adjacency matrix of the graph [11]. For clarity, v and u are vertices and vu is the edge with weight w_{vu} [11]. The paper aims to group the graph into clusters by looking at bubbles in the graph. There are three types of bubbles for $\overrightarrow{H_b}$ [11]. However, converging bubbles, where the connected edges are all incoming to the bubble, are the most important for establishing clusters as they are the end point of a directional path of the strongest connections [11]. Discrete clustering of the vertex set V(G) is achieved by "uniquely assigning each vertex to the converging bubble which is at the smallest shortest path distance" [11]. Firstly, vertices within converging bubbles are assigned based on whether they belong to only one bubble or multiple bubbles [11]. If a vertex belongs to one bubble, it is straightforwardly assigned to that bubble [11]. If it belongs to multiple bubbles, it is assigned to the bubble with the largest "strength" of attachment:

$$x(v, b_\alpha) = \frac{\sum_{u \in V(b_\alpha)} A_G(v,u)}{3(|V(b_\alpha)| - 2)} \tag{6}$$

where b_α is a converging bubble [11]. $|V(b_\alpha)|$ is the number of vertices in b_α with $3(|V(b_\alpha)| - 2)$ therefore equal to the number of edges [11]. Now that each converging bubble α has been assigned a unique set of vertices $V^0(\alpha)$, we can consider any remaining vertices outside converging bubbles. These vertices are located in the subtrees $\overrightarrow{h_\alpha}$ of the directed paths in $\overrightarrow{H_b}$ and contain the converging bubble b_α as its root [11]. However, a vertex may be a member of more than one subtree [11]. In such a case, we use a formula to work out the mean average shortest path distance given by

$$\overline{L}(v, \alpha) = \text{mean}\{l(v,u) \mid u \in V^0(\alpha) \wedge v \in V(\overrightarrow{h_a})\} \tag{7}$$

and assign the vertex to the converging bubble with the smallest distance [11]. This process results in a discrete partition of the vertex set V(G) into subsets associated with specific converging bubbles, forming distinct clusters within the network [11]. Discrete clustering is crucial as it makes it easier to analyse the characteristics of each cluster independently. A hierarchy is built at three levels using a specialised linkage procedure to explore each cluster's structure and how different clusters come together to form larger structures [11]. Complete linkage is a method used to measure the distance between clusters [11]. Utilising the method outlined already we have found that there exists a unique vertex set $V^\alpha(b_i)$ for every cluster α and bubble b_i in $\overrightarrow{h_a}$, where complete linkage is applied using the shortest path distances l(u,v) as the distance matrix to assess the intra-bubble hierarchy[11]. At the next stage up, the intra-cluster hierarchy is assessed using the distance matrix

$$d_a^I(b_i, b_j) = \max\{l(u,v) \mid u \in V^\alpha(b_i) \wedge v \in V^\alpha(b_j)\} \tag{8}$$

for the complete linkage procedure between the bubbles in $\overrightarrow{h_a}$ [11]. Finally, a complete linkage procedure is then applied between the clusters using a different distance matrix shown below [11]

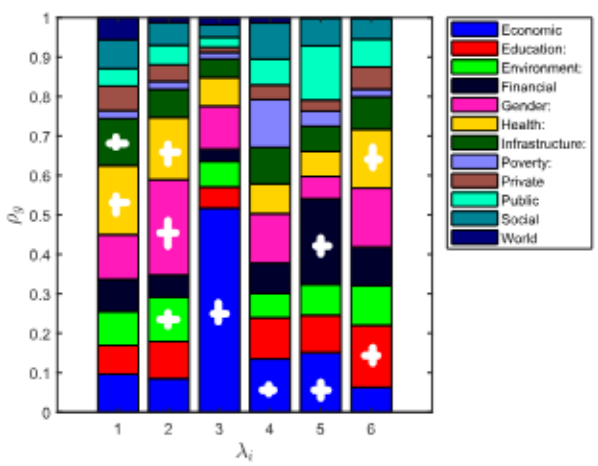
$$d^{II}(\alpha, \beta) = \max\{l(u,v) \mid u \in V(\alpha) \wedge v \in V(\beta)\}. \tag{9}$$

Applying this linkage procedure allows superclusters to be established and for clusters to be split into bubbles and their constituent elements below [11]. [11]. Dendrograms are a valuable tool to allow us to visually see the hierarchical organisation that is unveiled by the complete linkage procedure [11].

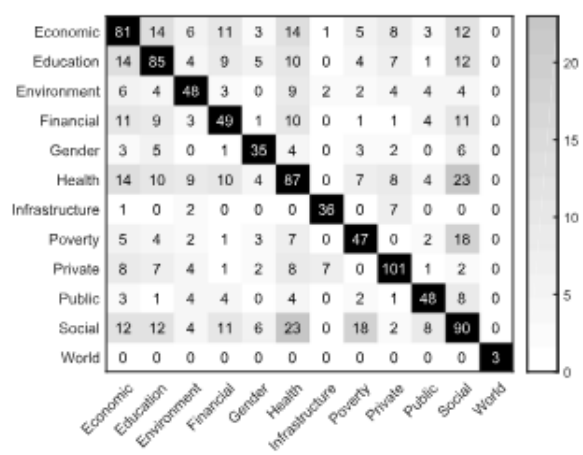
3 Results

The paper now looks at the most significant results of the paper by first looking at the results of the PCA. After applying PCA to the correlation matrix E, the Eigenvalue spectrum of the correlation matrix is analysed using RMT techniques. It is found that the eigenvalue distribution deviates from the Marčenko-Pastur (MP) distribution. Further research suggests that a nonrandom structure is apparent in the data [20]. Meanwhile, when the data is subsequently shuffled to destroy correlations between indicators, the fit with the MP distribution significantly improves [1]. As there is a structure in the data beyond randomness, discarding eigenvalues based on this mismatch may result in the loss of relevant information [20]. As a result, PCA may not be an appropriate method to apply to this dataset due to the loss of information.

The paper then investigates the structure of the data by plotting ρ_i for the top 6 principal components to reveal some interesting relationships between indicators captured by the principal components as shown by figure 1 [1]. The one-sided p-values for ρ_g, to test if a topic's contribution to the principal component is random, are calculated. The corresponding sections marked with a plus sign in Figure 1 are statistically significant (at the 5% significance level) where the null hypothesis is rejected [1]. The paper also plots the number of times two topics are simultaneously significant across all principal components to test whether the topics are interacting [1]. This is shown in Figure 3, where darker grey shades denote higher occurrences of simultaneous significance between topics across principal components [1]. The observed interactions between Social vs. Health and Social vs. Poverty highlight the complexity of the data's structure. This implies that no single topic overwhelmingly influences the data, and significant contributions stem from different topics in different combinations [1]. As the topics interact, it can be concluded that the structure of the topics does not mirror the actual structure between indicators [1]. This provides valuable insight for composite indicators designed to encapsulate specific aspects of development, such as the Human Development Index (HDI). Incorporating indicators that target different development facets could enhance the HDI's accuracy and effectiveness. Meanwhile, some indicators already included may provide no valuable information while adding unnecessary complexity. Despite revealing interesting structures in the dataset, it may be challenging to form composite indicators using PCA as the principal components only account for a limited part of the total variance, resulting in a loss of relevant information [21]. This means that potentially important indicators are ignored using PCA. This finding is further supported by other papers which find that while PCA is a powerful tool to reduce complexity and visualise data, it can give misleading information when constructing composite indicators [21].



(a) Figure 1: A bar chart od ρ_g for the top 6 principal of split by the 12 topics of the indicators. Source: [1]



(b) Figure 2: A grey scale map for the number of times two topics are statistically significant in the same principal component. Source: [1]

After applying the DBHT method to the dataset E to establish a partition of the indicators aligned with the data, the technique identifies a total of K = 102 clusters, each labelled from k = 1 to K [1]. The structure of these clusters is assessed using the preassigned topics in Figure 3, where the height of the bar corresponded to the number of individual indicators in each cluster N_k [1]. Substantially more clusters are detected when compared with the G topics, and the cluster sizes are highly diverse [1]. Despite this, it is found that each of the clusters has an economically intuitive meaning [1]. For example, even though cluster 72 is made up of a mixture of diverse topics, such as the Gini index and railway size, a closer inspection finds that all the measurements characterise underdeveloped countries [1]. Furthermore, cluster 21 contains C02 emissions, alternative and renewable energy, power distribution losses and more energy-related indicators [1]. Being able to interpret the the economic meaning behind the clusters highlights how well the clustering procedure has worked.

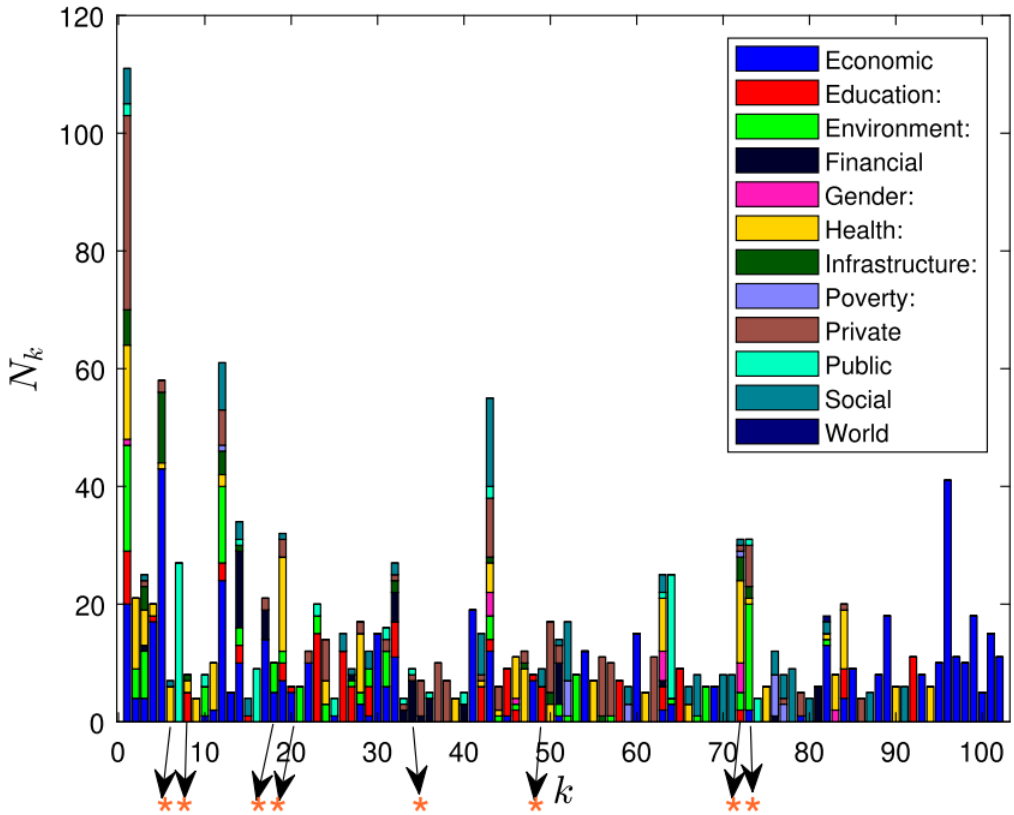


Figure 3: Distribution of topics amongst DBHT clusters. The cluster label k is plotted on the x axis against the number of indicators in each cluster Nk on the y axis. Each bar is broken down by the corresponding topic of the indicators contained in each. Topics shown by the key. Starting from the left clusters 6, 8, 18, 20, 34, 49, 72, 73, are marked by an orange asterisks and are discussed later. Source: [1].

The paper next quantitatively assesses how similar the clusters outputted by the DBHT are to the clusters based on the topics using the Adjusted Rand Index (ARI). This tool is used to evaluate the similarity between two different clusterings on the same data set [10]. The ARI equals 1 for identical clustering, -1 for an anti-correlation and 0 for independent clustering [10]. The paper calculates an ARI of 0.0456 when comparing the DBHT and topic clusterings, which further supports the paper's claim that the topics are not the best way to cluster the data [1]. However, questions have been raised regarding the significance of the ARI, primarily due to the strong assumptions it imposes on the distribution [22]. The analysis extends to a local level, examining if any topics exhibit significant presence within individual clusters [1]. After applying the method described in the paper, it is revealed that while many clusters

have one or two over-expressed topics, 49 clusters contain zero over-expressed topics [1]. Notably, these indicator clusters still possess economic significance, as exemplified by cluster 32, which encompasses education and infrastructure-related indicators linked to higher education. The clustering of indicators does not consistently align indicators of the same topic, meaning that indicators of the same topic are not necessarily correlated.

As the topics of the indicators do not accurately reflect the data structure, using the topics to aggregate a composite indicator may overlook crucial insights between the relationships of different indicator topics [1]. This paper aims to address this by proposing new data-driven composite development indicators leveraging on the clusters identified by DBHT [1]. Each cluster contains indicators of an economically interpretable topic that are correlated and share similar properties. Given the DBHT's ability to determine the number of clusters without prior input about the number of clusters, we have an intuitive way to reduce the subjectivity involved in aggregating composite indicators. The paper suggests using the set of clusters from the DBHT to select individual indicators that can effectively summarise their information. Therefore, it summarises information about the whole structure of the indicators and not a restricted subset.

The following text outlines the paper's method to calculate the new Cluster Driven Composite Indicators (CDCIs). The aim is to capture the average behaviour of all the indicators in each cluster k, by calculating the median value of all indicators for each cluster [1]. This is possible due to the standardisation and normalisation procedure used on the dataset; otherwise, the data would be incomparable. [1]. The composite indicator I_k of each cluster k is defined by the below formula

$$I_k = median_{i \in clusterk} X \tag{10}$$

where $i \in clusterk$ ensures the median is calculated for indicators, i , in cluster k. One key advantage of the median over other methods is that it is robust to outliers and skewed distributions [1]. However, a critique of the median is that it does not consider each indicator's relative importance. Although, the paper aims to avoid subjectivity and the relative importance may be difficult to interpret accurately if the weighted mean is used. A similar problem is also faced when using a PCA-based approach and calculating weights [21]. A further critique is that the median may not be appropriate for clusters with few observations. The smallest cluster contains just four observations, which is less than the rule of 5, the rule of thumb, and thus, it could be argued that this should be removed. However, with the average cluster size equal to 14.2, the approach used by the paper is relatively robust [1]. In total, this method gives a total of 102 indicators when applied to each cluster [1].

Pairs of CDCIs are analysed to assess a country's level of development. The paper specifically focuses on those marked by an asterisk in Figure 3, although this paper will look at a subset of these comparisons between indicators to highlight the key findings. The first key finding is that some plots exhibit a "hockey-stick" shape, as shown by Figure 4 [1]. When looking at the particular countries in detail, it is found that developing countries tend to make up the vertical leg while more developed countries make up the horizontal leg [1]. The paper notes that there appears to be a saturation effect for developed countries [1]. This paper interprets that this may indicate diminishing returns where countries may exhibit fast development initially but need help to see development at the same pace past a certain point. The existence of this pattern also indicates a clear transition from developing to developed nations, supporting the two-regime hypothesis [1]. This transition can be seen across time in the middle panel of Figure 4 [1]. Using the hockey stick shape of certain CDCI pairs, the paper can initially distinguish between developing and developed countries but find that the pairs lose discriminatory power as countries progress [1]. For example, in the top panel where $I_3 4$ (use of mobile and banking services) is plotted against $I_4 9$ (secondary school enrollment) [1]. It is found that for developing countries, the use of mobile phone technology can signify their stage of development [1]. However, due to the saturation of mobile phone technology in

developed countries, it may be unreliable to distinguish between their level of development [1]. The relationships of some pairs of CDCIs do not follow the hockey-stick shape, such as natural resource abundance I_18 vs. underdevelopment I_72. The relationship between these 2 two CDCIs is characterised by the resource curse [1]. This is where countries often use up most of their resources during their development, and developed countries often have very few resources [1].

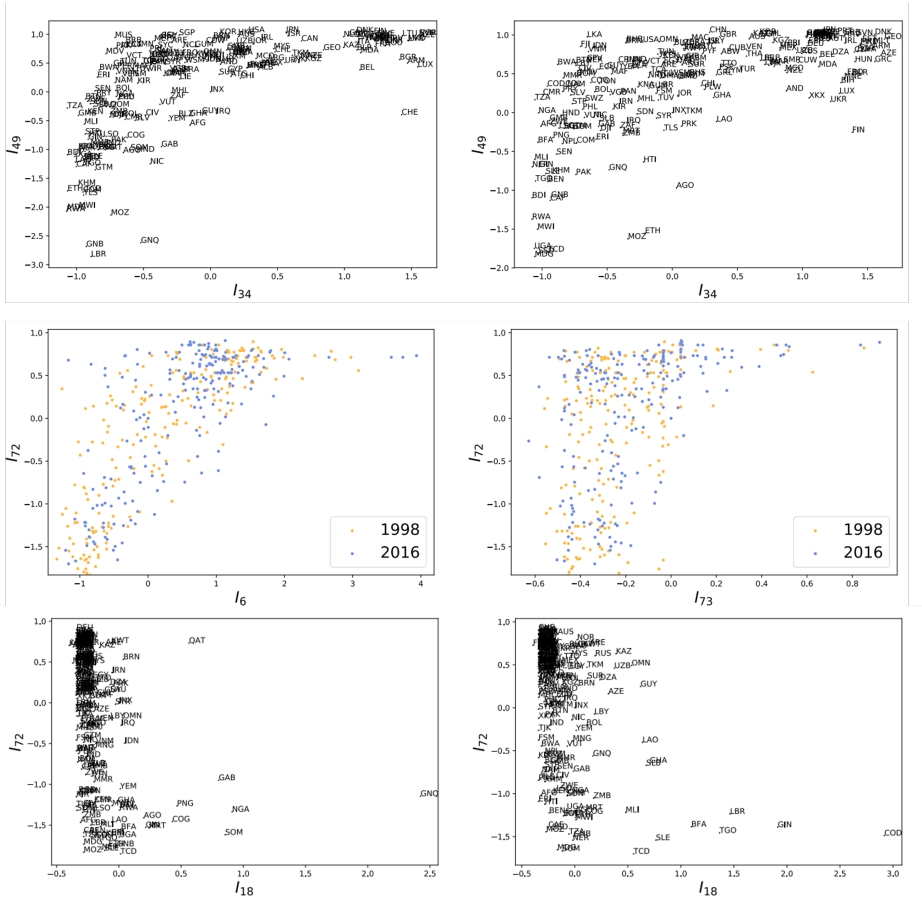


Figure 4: Top Panel - Comparison between I34 vs I49 with hockey stick shape for 1998 (left) and 2016 (right). Middle Panel - Comparison between differed CDCIs with hockey stick shape. On the left - I6 vs I72 for 1998 in orange and 2016 in blue. On the right is the same but for I73 vs I72. Bottom Panel - Comparison between I18 vs I72 with no hockey-stick shape for 1998 (left) and 2016 (right). Source: [1].

As previously mentioned, the connectivity of a node is the number of edges incoming or outgoing from the node [17]. The most connected nodes are known as hubs, and they play a crucial role in the structure and function of networks [23]. Therefore, the paper suggests that these nodes could be highly influential in analysing the development of countries [23]. Identifying these nodes offers a potential solution for forming a reliable and reduced set of development indicators [1]. This subset could avoid issues from calculating composite versions of indicators mentioned previously [1]. The PMFG method is applied to the correlation matrix E to identify the most critical indicators. The network structure for all the indicators can be visualised in Figure 5. Centrality is a concept that evaluates the importance of nodes within a network, and there exist many network centrality measures to identify these influential indicators [1]. One example is degree centrality, a straightforward measure that counts the number of connections a node has [23]. However, this ignores the quality or strength of connections [23]. Instead, the paper opts for PageRank, which assesses a node's importance based on both the quantity and quality of incoming connections, as the quality of connections matters [1]. The paper finds that Pagerank can discern common metrics used in development indicators, like GDP and government debt [1]. However, it surprisingly identifies the top

indicator as mobile cellular subscriptions, which is over 2.5 times more important than GDP [1]. The paper attributes the robustness of mobile subscriptions as an indicator of their ability to encapsulate various development facets. The opposite is true for the third most important indicator, the % of male children workers aged 7-14 [1]. This encapsulates many potential avenues of underdevelopment, such as a lack of an education system, high poverty levels and low child rights and social development. The problem of child labour is closely associated with poverty and underdevelopment and is particularly prevalent in Asia and Africa[24]. The paper also finds that no single topic is more influential than others [1]. However, the size of the subset must still be subjectively chosen.

Figure 5: The cluster allocation of the development indicators visualised on the PMFG network by the colour of the vertices. Source: supplementary for [1].

A performance comparison can be conducted between CDCIs, PageRank and a random benchmark by leveraging on the fact that the set of indicators should be able approximately reconstruct the original data, as they capture primary elements shaping the correlations [1]. The paper proposes a linear factor model representing each original indicator X_i as a linear combination of composite indicators $\tilde{I}_k$, where β_{ik} represents the sensitivity of X_i to changes in $\tilde{I}_k$ [1]. In the formula

$$X_i = \sum_{k=1}^{K} \beta_{ik} \tilde{I}_k + \in_i \tag{11}$$

$\in_i$ is a white noise term [1]. The number of indicators is set to K for all methods, as the number of indicators has an impact on the ability to describe the correlations [1]. Elastic net regression is used and the performance is assessed based on the error between the linear model in Equation (11) and the real indicator values, using mean squared error (MSE) and error reduction ratio (ERR) metrics [1]. The MSE measures the average squared difference between the predicted values $X_i^{(predict)}$ based on the elastic regression model and the actual values of the indicators X_i [1]. It's computed as

$$MSE = \sum_{i=1}^{N} (X_i^{(predict)} - X_i)^2. \tag{12}$$

The ERR is calculated as the ratio of the MSE_CDCIs for CDCIs to the MSE_alt for alternative schemes of composite indicators. It's calculated using equation as

$$ERR = \frac{MSE_{CDCIs}}{MSE_{alt}}. \tag{13}$$

If ERR is less (greater) than 1, it indicates that CDCIs perform better (worse) than the alternative schemes [1]. The two alternatives that are compared are the top K PageRank indicators and the average of 100 randomly selected subsets of K indicators from the dataset [1]. The random set provides a benchmark for the performance of CDCIs and PageRank. The results, as presented in Table 2, show that ERR is much less than 1 in both cases, indicating that CDCIs outperform both the random benchmark and PageRank [1]. An ERR value of 0.66 when compared with the random benchmark demonstrates a notable improvement in performance by the CDCIs [1].Therefore, it can be concluded that CDCIs are more effective at capturing the essential structure of correlations while reducing the dimensionality of the dataset compared to the alternatives [1]. It would however be interesting to see the results of taking the indicator with the highest page rank score for each cluster to see how a combination of the two methods would perform. This would solve both issues of subjectivity and calculating composite versions of indicator. Nevertheless, the scope of the results is significant given the large amount of research done on composite development indicators and their vital role in informing policy decisions. This paper believes this methodology should be applied to other datasets to compare the efficiency of current composite development indicators and be put into practice where possible.

Random	PageRank
0.66	0.71

Table 2: The ERR calculated by equation (13) for the average of 100 subsets of 102 indicators in the first column and 102 of the most influential indicators using PageRank in the second column. Source: [1].

The paper also briefly looks at the dynamic stability of clusters, which is done by comparing the similarity between selected time windows using the ARI [1]. An ARI of 0.376 suggests a high level of similarity between the DBHT clusterings of indicators [1]. The same method is applied to investigate the dynamic stability of the relationship between DBHT clusterings of CDCIs and finds an even higher ARI value of 0.683 [1]. This analysis indicates that the clusters derived from both individual indicators and CDCIs demonstrate a consistent and stable structure over time, providing valuable insights into the dynamics of the dataset.

This paper's most significant criticism of the methods applied is due to the use of Pearson's mean correlation coefficient (PMCC). The PMCC, although quick and easy to compute, is only a measure of the linear relationship between two variables. This assumption of linearity means that in nonlinear cases, entirely dependent variables can have a zero correlation coefficient. Also, while measures have been implemented to change the distribution of indicators to allow for comparability of indicators, their application may result in intricate or nuanced relationships within the data that may be lost or smoothed out. Therefore, how do we best measure dependency? One possibility is to use rank correlation, which measures a monotonic association between two variables [25]. Rank Correlation may be beneficial given its works for non-normally distributed data, which is commonplace in financial data [14]. Another possibility is to use a multiplex approach, which can simultaneously analyse different kinds of dependencies among financial time series [26]. A similar methodology could be applied to cross-sectional data. The multiplex approach allows for an investigation of the interplay between linear, nonlinear, and Tail dependencies by looking at four dependency measures, namely Pearson, Kendall, Tail, and Partial [26]. Therefore, it may reveal features not revealed by analysing each dependency in isolation [26]. However, both of these methods add complexity, and the need for comparability between indicators must be maintained for the effort to

be worth the reward. Overall, despite the assumption of linearity and the possibility that correlations between indicators may not consistently reflect the true causal influence between them, CDCIs offer an advantage over current methods of constructing composite development indicators [2].

4 Conclusion

The paper first made a case for the need for an improvement in the process of selecting what and how indicators should be aggregated to form composite development indicators. A total of N = 1448 indicators were initially divided into G = 12 topics by their root objective [1]. Through the application of PCA and the combination of network filtering techniques and hierarchical clustering, it has been determined that the structure of the topics does not accurately reflect the actual relationships between indicators [1]. As well as this, PCA may discard valuable information [1]. The paper then sought to employ network filtering techniques and hierarchical clustering to overcome some of the critical problems with existing techniques, such as how and what indicators to aggregate [1]. The paper finds that the indicator clusters found after applying the DBHT contain a mixture of topics but are economically intuitive [1]. These results are used to propose a new set of cluster-driven composite development indicators that are interpretable, objective, and data-driven [1]. Another method to form a composite development indicator using PageRank is also proposed [1]. A financial interpretation of some key results from CDCIs and PageRank is then detailed. Finally, the performance of both methods is compared against a random benchmark, which finds that CDCIs outperform the random benchmark and PageRank [1].

Even though this paper has a few criticisms of the methodology, other than testing the combination of PageRank and CDCIs, the suggestions for improvement add additional complexity. If this level of complexity can be handled, these suggestions may offer a greater insight into the structure between indicators. However, this paper believes that the current methodology to form CDCIs proposed in the paper already offers a distinct advantage over current methods in the literature. This paper proposes that someone from the relevant field should compare the efficiency of current composite development indicators and put CDCIs into practice where possible.

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